

MENGSHAN, China

WMO: 590580

Lat: 24.2036N

Lon: 110.5109E

Elev: 601

StdP: 14.38

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.4	38.8	23.7	18.3	49.4	28.0	22.4	47.2	13.9	44.7	12.3	47.1	5.2	320	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	94.3	78.2	92.6	78.1	91.0	78.0	81.0	89.2	80.4	88.2	79.7	87.2	4.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.0	154.1	84.9	78.3	150.3	84.1	77.7	147.5	83.4	45.5	89.6	44.7	88.4	43.9	87.6	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.8	10.0	9.1	33.0	98.3	2.0	1.7	31.5	99.5	30.3	100.5	29.2	101.4	27.7	102.7
			31.5	82.0	2.3	1.2	29.8	82.8	28.4	83.5	27.1	84.2	25.5	85.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	69.3	50.8	54.6	60.5	69.9	77.0	80.9	83.1	83.1	79.6	73.2	64.0	54.6	(6)
		DBStd	12.85	7.51	9.01	7.54	6.70	4.48	3.27	2.80	2.97	4.25	5.24	6.27	6.14	(7)
		HDD50	161	83	47	7	0	0	0	0	0	0	2	22		(8)
		HDD65	1389	443	309	180	31	0	0	0	0	7	90	329		(9)
		CDD50	7220	107	176	333	598	836	926	1026	888	719	422	163		(10)
		CDD65	2974	2	18	41	179	371	476	561	561	438	261	60	6	(11)
		CDH74	26390	5	71	140	955	2782	4457	6038	5967	3939	1712	311	13	(12)
		CDH80	9379	0	8	14	208	820	1466	2412	2436	1474	499	42	0	(13)
(14)	Wind	WSAvg	3.8	4.7	4.5	3.9	3.5	3.3	3.1	3.1	3.1	3.5	3.9	4.1	4.7	(14)
(15)	Precipitation	PrecAvg	66.6	2.0	3.1	3.8	8.2	11.5	10.6	7.9	8.1	4.0	3.4	2.4	1.6	(15)
(16)		PrecMax	99.6	5.6	9.3	8.6	17.1	23.9	20.9	22.9	16.5	8.4	8.7	6.8	8.2	(16)
(17)		PrecMin	45.2	0.1	0.2	1.1	2.6	3.7	1.4	2.4	0.2	0.4	0.2	0.0	0.0	(17)
(18)		PrecStd	12.3	1.5	2.3	1.9	3.1	4.4	5.2	5.2	4.0	2.0	2.6	2.0	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	74.2	80.7	81.5	88.4	92.1	93.6	97.1	96.8	94.5	90.9	84.5	75.6		(19)
(20)		0.4% MCWB	64.1	68.3	70.8	75.1	76.9	78.6	78.4	78.3	77.4	73.9	70.3	65.6		(20)
(21)		2% DB	69.8	75.9	77.6	85.2	89.5	91.4	94.2	94.3	92.1	87.8	80.8	71.9		(21)
(22)		2% MCWB	60.8	65.9	68.5	73.8	76.3	78.8	78.6	78.3	76.9	72.4	67.0	61.4		(22)
(23)		5% DB	66.1	71.7	74.5	82.3	87.3	89.6	92.2	92.5	90.3	85.2	77.8	68.4		(23)
(24)		5% MCWB	58.7	64.5	67.2	73.0	75.8	78.4	78.8	78.3	76.7	71.5	66.0	58.6		(24)
(25)		10% DB	62.6	68.0	71.3	79.5	84.8	87.5	90.2	90.5	88.1	82.8	74.5	64.6		(25)
(26)		10% MCWB	56.9	62.7	65.8	72.1	75.6	78.2	78.7	78.4	76.1	70.4	64.7	56.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	66.1	70.0	73.2	78.0	80.6	81.7	82.1	81.9	80.7	77.6	73.6	70.1		(27)
(28)		0.4% MCDB	71.6	78.9	79.3	84.5	87.3	89.2	91.0	90.5	89.3	85.9	79.8	72.9		(28)
(29)		2% WB	63.3	67.5	71.2	76.2	79.2	80.6	81.0	80.9	79.5	76.2	71.6	65.0		(29)
(30)		2% MCDB	68.0	73.2	75.9	82.3	85.7	87.8	89.3	89.4	87.6	83.3	76.8	68.7		(30)
(31)		5% WB	60.6	65.4	69.0	74.8	78.1	79.8	80.1	80.1	78.7	74.9	69.6	60.9		(31)
(32)		5% MCDB	64.6	70.2	72.6	80.1	84.2	86.5	88.0	88.2	86.2	80.9	74.8	66.0		(32)
(33)		10% WB	57.2	63.2	66.9	73.5	77.0	79.0	79.4	79.3	77.8	73.3	67.0	57.7		(33)
(34)		10% MCDB	61.3	67.9	70.4	77.9	82.7	85.2	86.8	86.9	84.6	78.7	72.0	62.9		(34)
(35)	Mean Daily Temperature Range	MDBR	10.9	10.7	10.3	10.8	12.0	10.9	12.7	13.3	14.1	15.0	14.8	13.7		(35)
(36)		5% DB MCDBR	20.3	18.0	17.0	16.0	16.6	14.9	16.6	17.1	17.7	19.1	20.7	21.8		(36)
(37)		5% WB MCWBR	11.0	9.0	8.4	6.8	6.0	4.9	4.4	4.6	5.2	6.5	8.3	11.0		(37)
(38)		5% WB MCDBR	15.3	14.1	12.1	12.3	12.2	12.0	14.0	14.8	14.5	14.3	13.9	15.2		(38)
(39)	Clear-Sky Solar Irradiance	taub	0.645	0.692	0.830	0.795	0.633	0.594	0.565	0.637	0.679	0.724	0.613	0.628		(40)
(41)		taud	1.602	1.563	1.370	1.467	1.801	1.911	1.984	1.778	1.669	1.539	1.713	1.656		(41)
(42)		Ebn at Noon	193	197	179	191	225	232	239	222	208	187	200	191		(42)
(43)		Edh at Noon	75	84	106	98	70	62	58	71	77	84	67	69		(43)
(44)	All-Sky Solar Radiation	RadAvg	644	693	761	961	1194	1243	1515	1489	1364	1216	936	780		(44)
(45)		RadStd	162	148	129	135	124	178	150	100	111	114	160	174		(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	+0.76	N/A	N/A	N/A	N/A	+158	+117	(47)

Nomenclature: See separate page